AI-GENERATED ENGINEERING PROTOTYPE — DAB-V1 — HUMAN REVIEW REQUIRED — NOT HARDWARE QUALIFIED

AIPE 10 kW dual active bridge, 800 V primary / 400 V secondary, 100 kHz, single-phase-shift; generation DAB-V1.
Requirement files: examples/dab-10kw/requirements; insulation table values are reference inputs to be reviewed.
No fabrication, energisation or procurement is authorised by this schematic.

Full bridge, DC link, switch-node terminals, DC-blocking capacitors and transformer current sensor



文件: 01_bridge.kicad_sch

Leg A high side: isolated gate driver, DESAT, gate loop and isolated bias



文件: 02_drive_ah.kicad_sch

Leg A low side: isolated gate driver, DESAT, gate loop and isolated bias



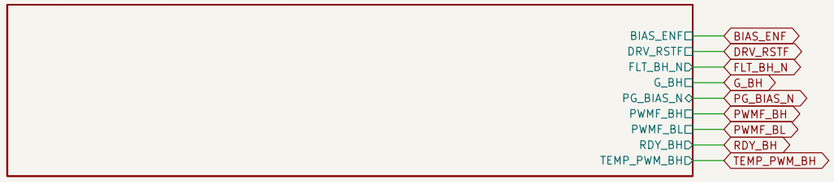
文件: 03_drive_al.kicad_sch

Leg B low side: isolated gate driver, DESAT, gate loop and isolated bias



文件: 04_drive_bl.kicad_sch

Leg B high side: isolated gate driver, DESAT, gate loop and isolated bias



文件: 05_drive_bh.kicad_sch

Isolated bus-voltage sensing, isolated sensing supply, Hall-sensor conditioning and temperature



文件: 06_sense.kicad_sch

Keyed control harness, filtered inputs, auxiliary rails and mounting



文件: 07_interface.kicad_sch

AI-GENERATED ENGINEERING PROTOTYPE — DAB-V1 — HUMAN REVIEW REQUIRED — NOT HARDWARE QUA

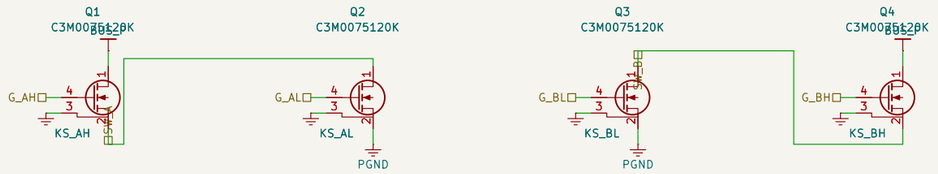
Sheet: /
File: dab-primary.kicad_sch

Title: AIPE 10 kW DAB — primary power board (800 V bus)

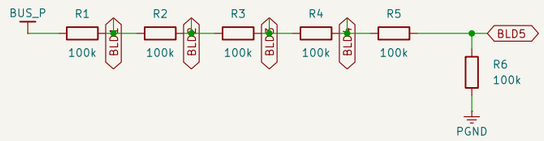
Size: A3	Date:	**Rev: DAB-V1**
KiCad E.D.A. 10.0.6		Id: 1/8

Full bridge, DC link, switch-node terminals, DC-blocking capacitors and transformer current sensor

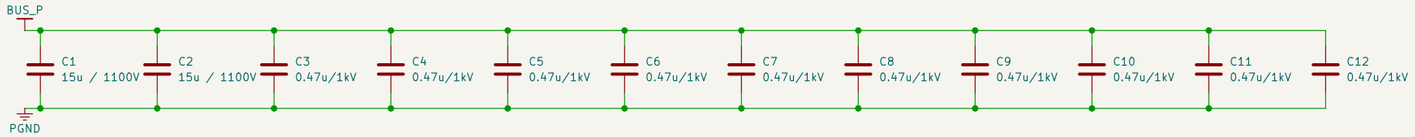
primary full bridge: 4x C3M0075120K with Kelvin source (TO-247-4)



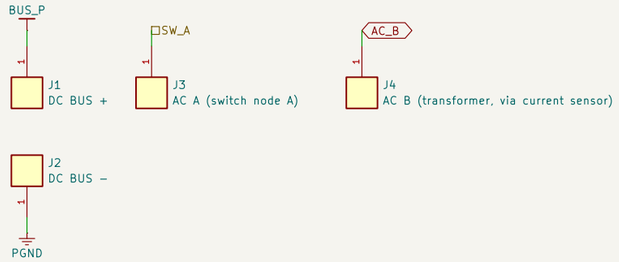
Passive DC-link discharge chain (independent of auxiliary power)



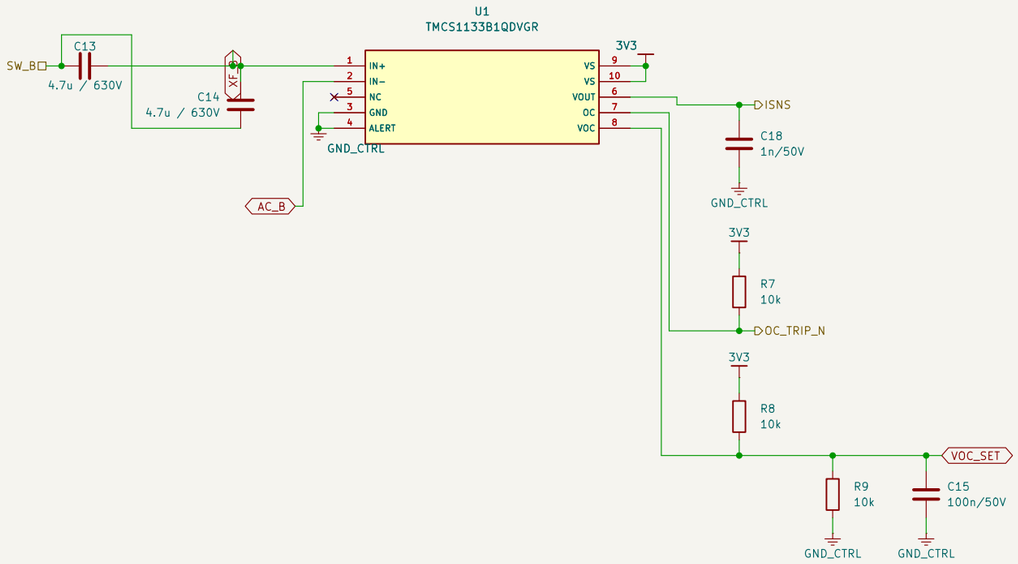
primary DC link: film bulk plus distributed HV ceramics at each leg (KR-CAP-003)



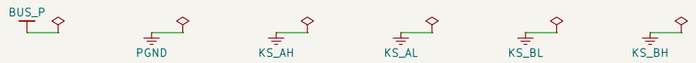
Press-in HV terminals: DC bus and the two AC nodes



Leg B AC path: DC-blocking capacitors and transformer current sensor



Supply provenance (PWR_FLAG)



AI-GENERATED ENGINEERING PROTOTYPE – DAB-V1 – HUMAN REVIEW REQUIRED – NOT HARDWARE QUA

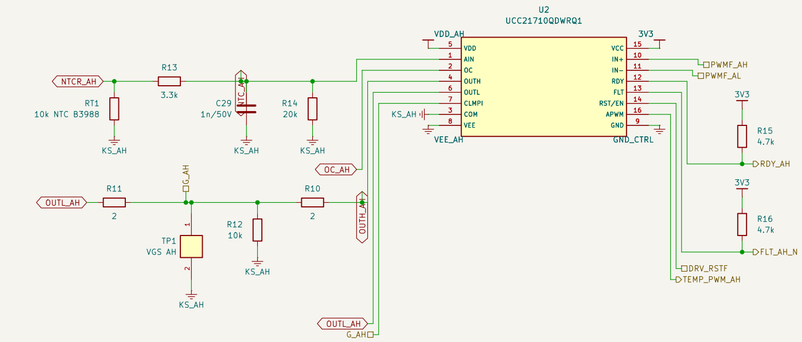
Sheet: /Full bridge, DC link, switch-node terminals, DC-blocking capacitors and transformer current s
File: 01_bridge.kicad_sch

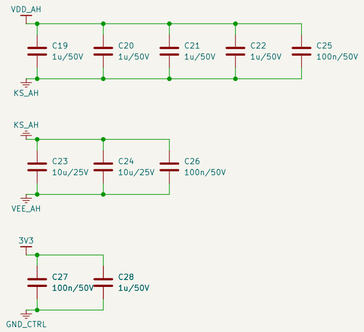
Title: Full bridge, DC link, switch-node terminals, DC-blocking capacitors ar

Size: A3	Date:	Rev: DAB-V1
KiCad E.D.A. 10.0.6		Id: 2/8

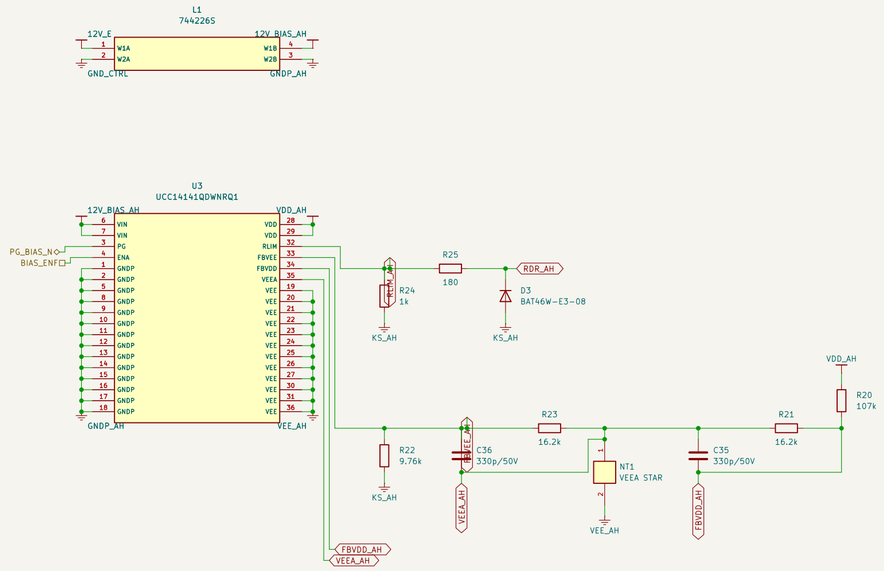
Leg A high side: isolated gate driver, DESAT, gate loop and isolated bias

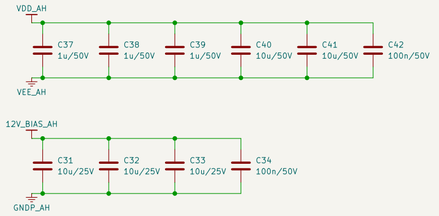
AH: isolated gate driver, Kelvin return, Miller clamp and gate loop





AH: isolated DC/DC bias module, input filter and quiet feedback star

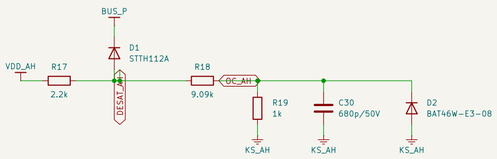




Supply provenance (PWR_FLAG)



AH: drain-voltage short-circuit detection



AI-GENERATED ENGINEERING PROTOTYPE – DAB-V1 – HUMAN REVIEW REQUIRED – NOT HARDWARE QUA

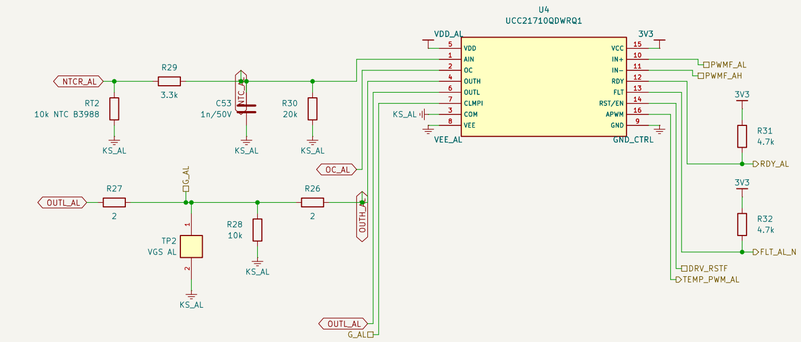
Sheet: /Leg A high side: isolated gate driver, DESAT, gate loop and isolated bias/
File: 02_drive_ah.kicad_sch

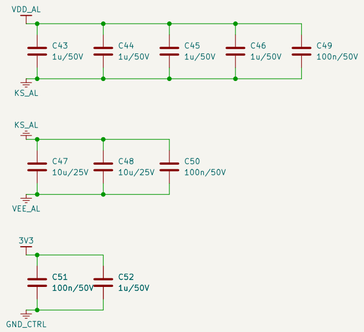
Title: Leg A high side: isolated gate driver, DESAT, gate loop and isolated bia

Size: A2	Date:	Rev: DAB-V1
KiCad E.D.A. 10.0.6		Id: 3/8

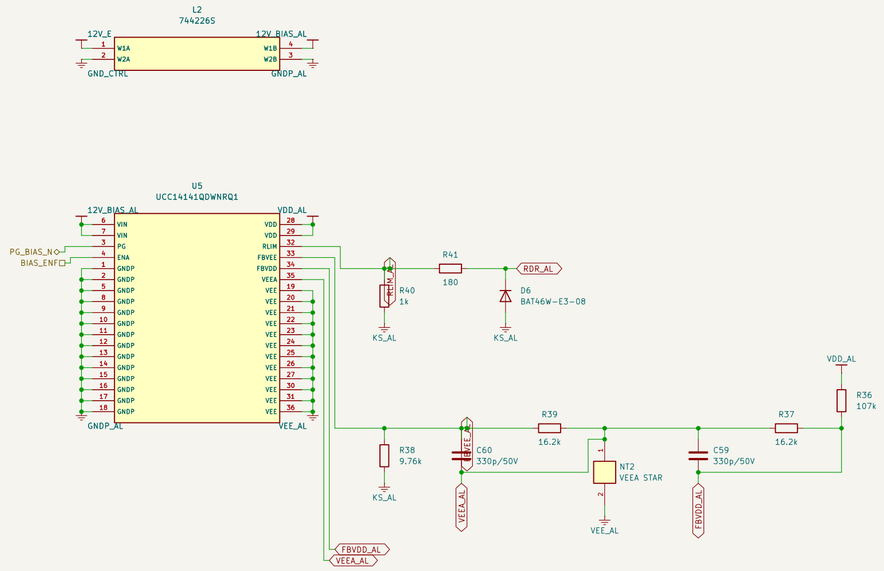
Leg A low side: isolated gate driver, DESAT, gate loop and isolated bias

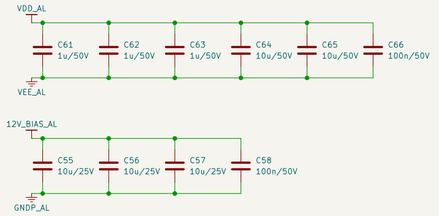
AL: isolated gate driver, Kelvin return, Miller clamp and gate loop



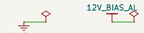


AL: isolated DC/DC bias module, input filter and quiet feedback star

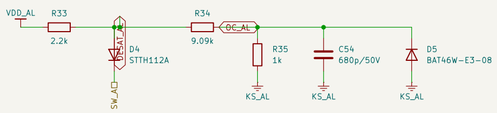




Supply provenance (PWR_FLAG)



AL: drain-voltage short-circuit detection



AI-GENERATED ENGINEERING PROTOTYPE — DAB-V1 — HUMAN REVIEW REQUIRED — NOT HARDWARE QUA

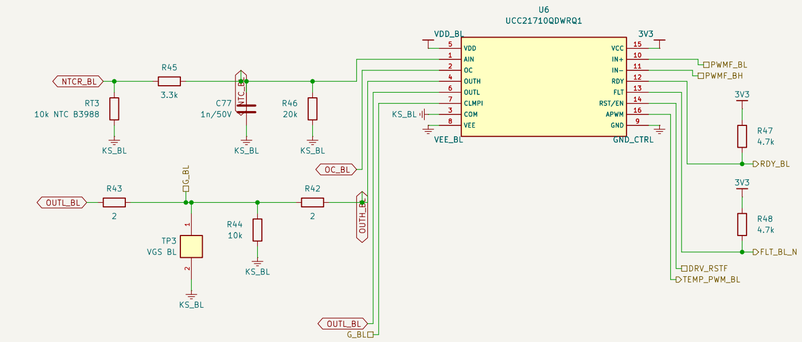
Sheet: /Leg A low side: isolated gate driver, DESAT, gate loop and isolated bias/
File: 03_drive_al.kicad_sch

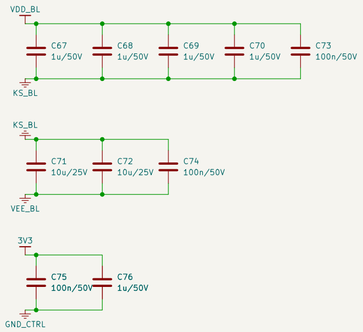
Title: Leg A low side: isolated gate driver, DESAT, gate loop and isolated bia

Size: A2	Date:	Rev: DAB-V1
KiCad E.D.A. 10.0.6		Id: 4/8

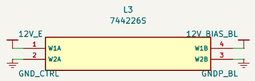
Leg B low side: isolated gate driver, DESAT, gate loop and isolated bias

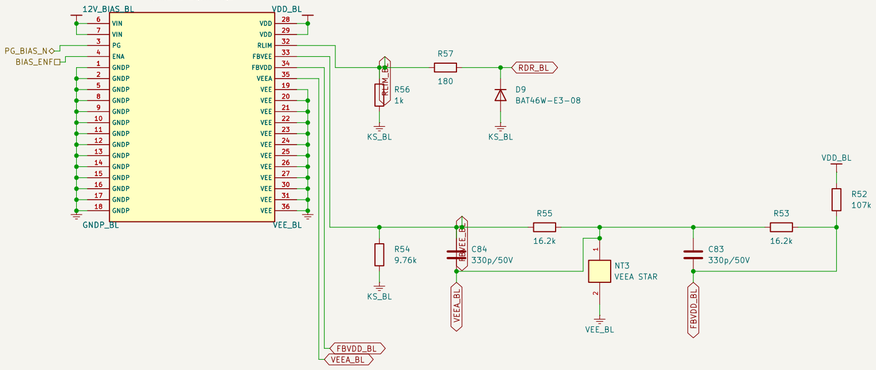
BL: isolated gate driver, Kelvin return, Miller clamp and gate loop

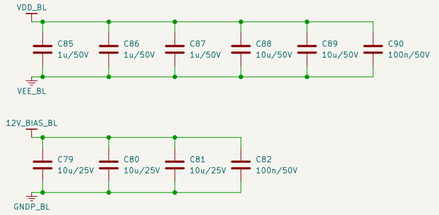




BL: isolated DC/DC bias module, input filter and quiet feedback star



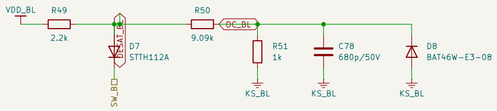




Supply provenance (PWR_FLAG)



BL: drain-voltage short-circuit detection



AI-GENERATED ENGINEERING PROTOTYPE — DAB-V1 — HUMAN REVIEW REQUIRED — NOT HARDWARE QUA

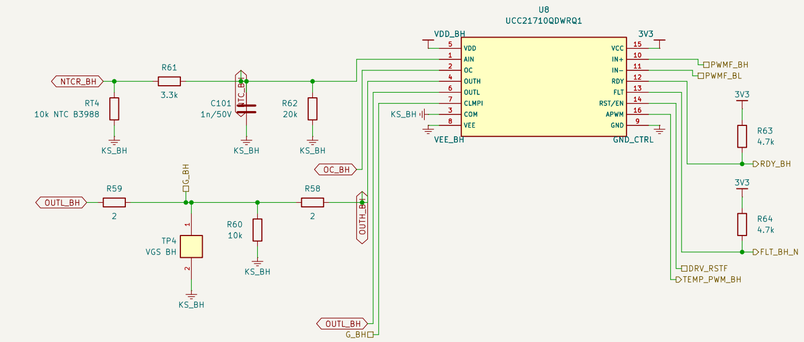
Sheet: /Leg B low side: isolated gate driver, DESAT, gate loop and isolated bias/
File: 04_drive_bl.kicad_sch

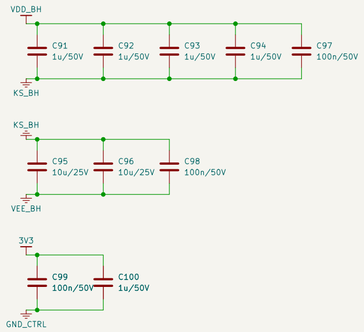
Title: Leg B low side: isolated gate driver, DESAT, gate loop and isolated bia

Size: A2	Date:	Rev: DAB-V1
KiCad E.D.A. 10.0.6		Id: 5/8

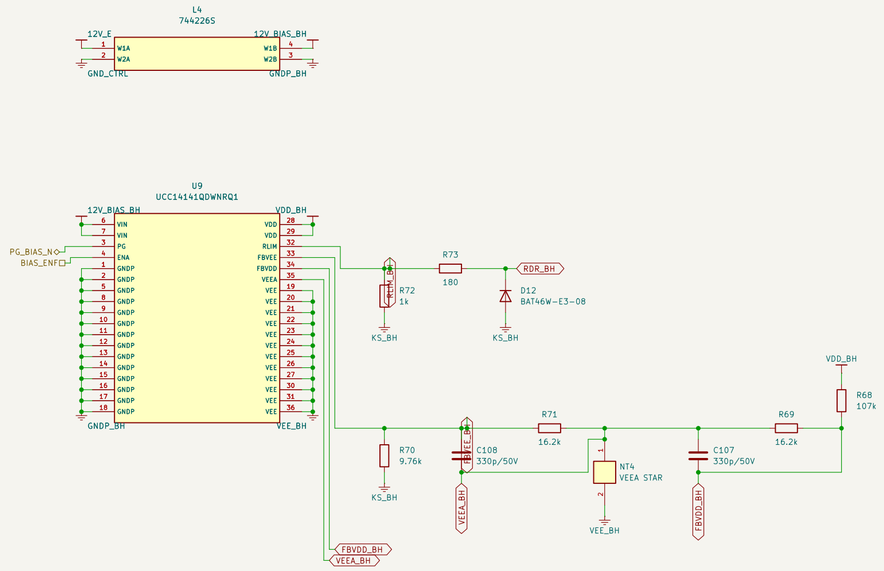
Leg B high side: isolated gate driver, DESAT, gate loop and isolated bias

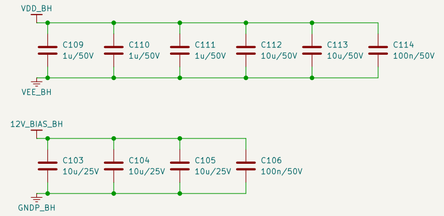
BH: isolated gate driver, Kelvin return, Miller clamp and gate loop





BH: isolated DC/DC bias module, input filter and quiet feedback star

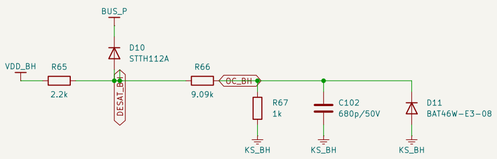




Supply provenance (PWR_FLAG)



BH: drain-voltage short-circuit detection



AI-GENERATED ENGINEERING PROTOTYPE — DAB-V1 — HUMAN REVIEW REQUIRED — NOT HARDWARE QUA

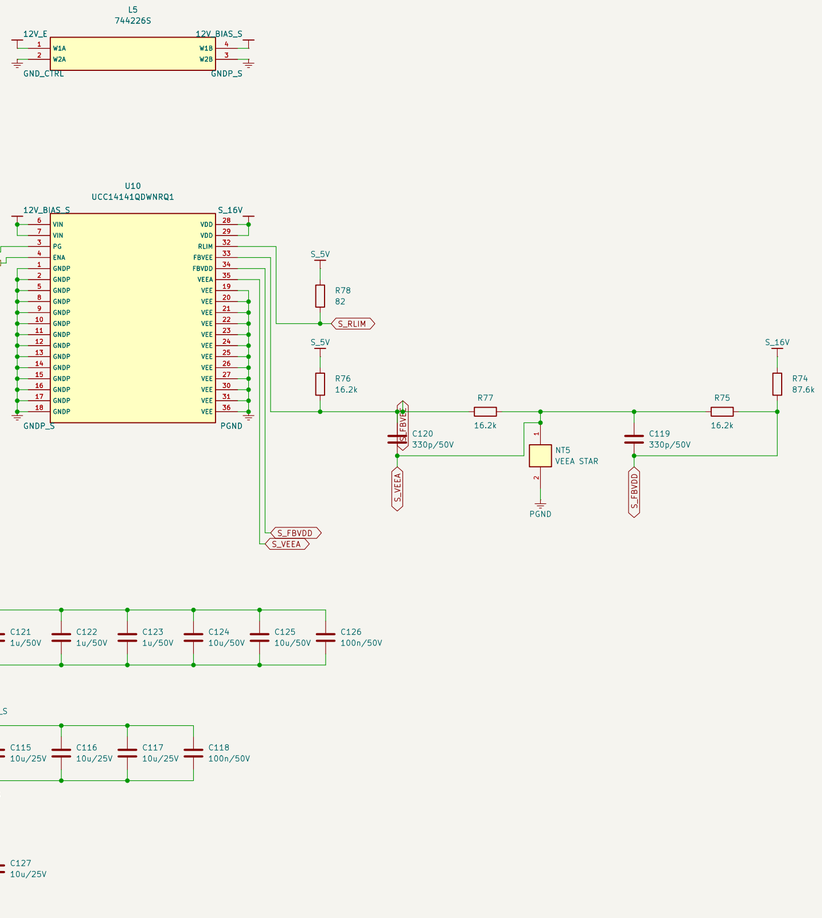
Sheet: /Leg B high side: isolated gate driver, DESAT, gate loop and isolated bias/
File: 05_drive_bh.kicad_sch

Title: *Leg B high side: isolated gate driver, DESAT, gate loop and isolated bi*

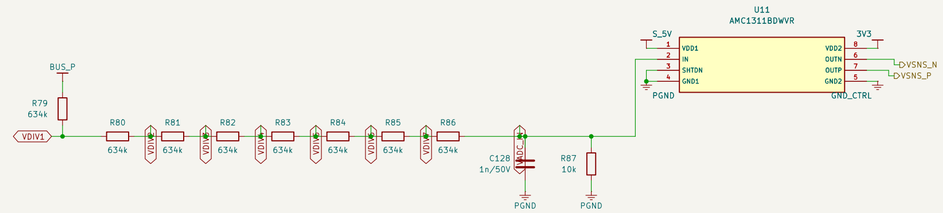
Size: A2	Date:	**Rev: DAB-V1**
KiCad E.D.A. 10.0.6		Id: 6/8

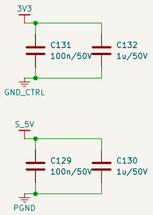
Isolated bus-voltage sensing, isolated sensing supply, Hall-sensor conditioning and temperature

S: isolated DC/DC bias module, input filter and quiet feedback star

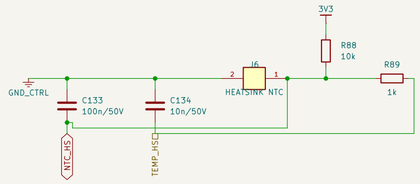


primary bus voltage: high-impedance ladder into the isolated amplifier





Heat-sink NTC (cabled, SELV) and harness filter



Supply provenance (PWR_FLAG)



AI-GENERATED ENGINEERING PROTOTYPE — DAB-V1 — HUMAN REVIEW REQUIRED — NOT HARDWARE QUA

Sheet: /Isolated bus-voltage sensing, isolated sensing supply, Hall-sensor conditioning and temperat
File: 06_sense.kicad_sch

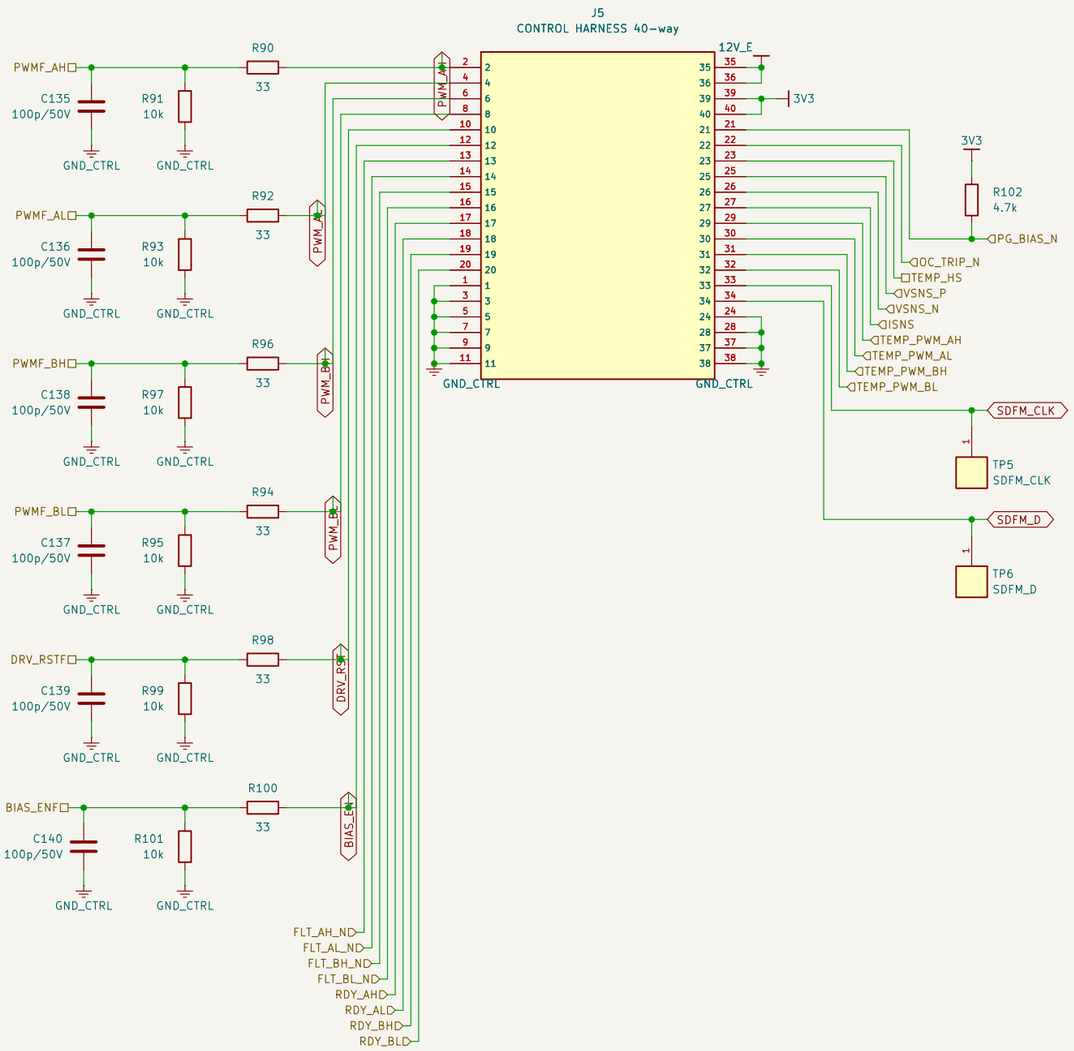
Title: Isolated bus-voltage sensing, isolated sensing supply, Hall-sensor co

Size: A2	Date:	Rev: DAB-V1
KiCad E.D.A. 10.0.6		Id: 7/8

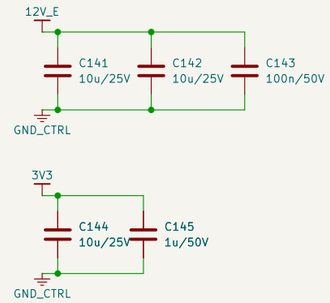
Keyed control harness, filtered inputs, auxiliary rails and mounting

PWR_FLAG symbols state supply provenance for ERC; they do not prove voltage, current capacity or protection.

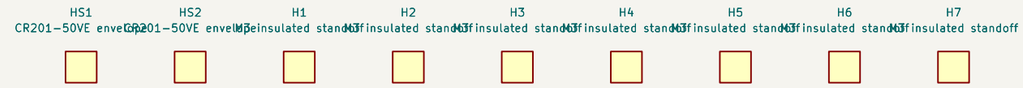
Keyed SELV control harness (40-way) and filtered logic inputs



Auxiliary rails from the harness



Heat-sink envelopes and insulated mounting



Supply provenance (PWR_FLAG)



AI-GENERATED ENGINEERING PROTOTYPE — DAB-V1 — HUMAN REVIEW REQUIRED — NOT HARDWARE QUA

Sheet: /Keyed control harness, filtered inputs, auxiliary rails and mounting/
File: 07_interface.kicad_sch

Title: Keyed control harness, filtered inputs, auxiliary rails and mounting

Size: A3	Date:	**Rev: DAB-V1**
KiCad E.D.A. 10.0.6		Id: 8/8